

GFAz: analyze pangenome graphs without unpacking them

A lossless, structure-aware GFA compressor whose traversal grammar doubles as a streaming compute substrate.

Taolue Yang · Youyuan Liu · Bo Jiang · Xinghua Shi · Sian Jin — Temple University

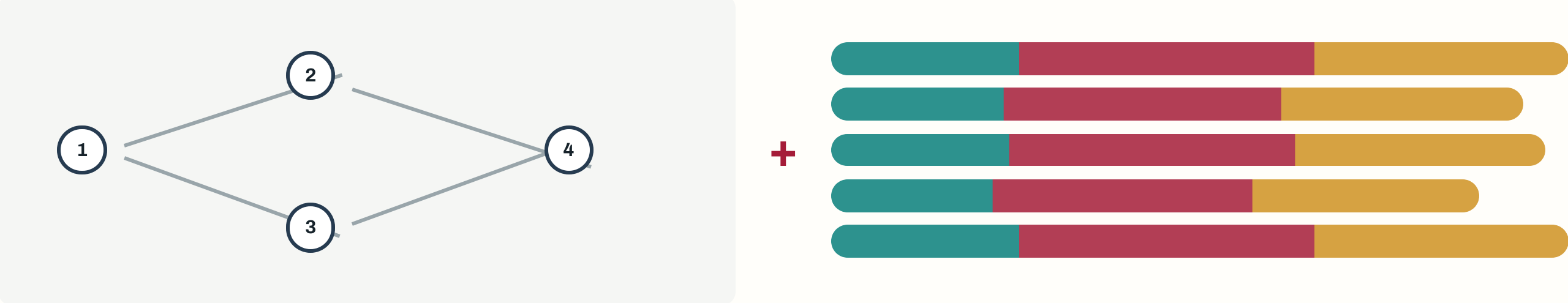
369 GB → 4.5 GB

HPRC v2.1 becomes 82.8× smaller while retaining full GFA semantics

Compress the repeated walks once. Then stream them directly into decompression or analysis.

01 The scaling problem is in the walks

Pangenome topology grows slowly; traversal records grow with every haplotype.

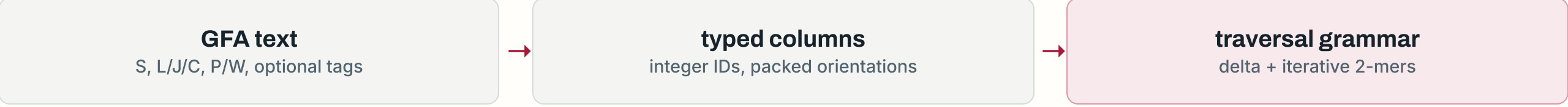


Across human graphs, P/W records dominate the bytes. Closely related haplotypes repeat long node-ID substrings, but gzip and Zstd see only short byte windows.

>90%
of large GFAs can be traversal records

linear growth
as samples and haplotypes are added

02 Compress the structure, not just the text



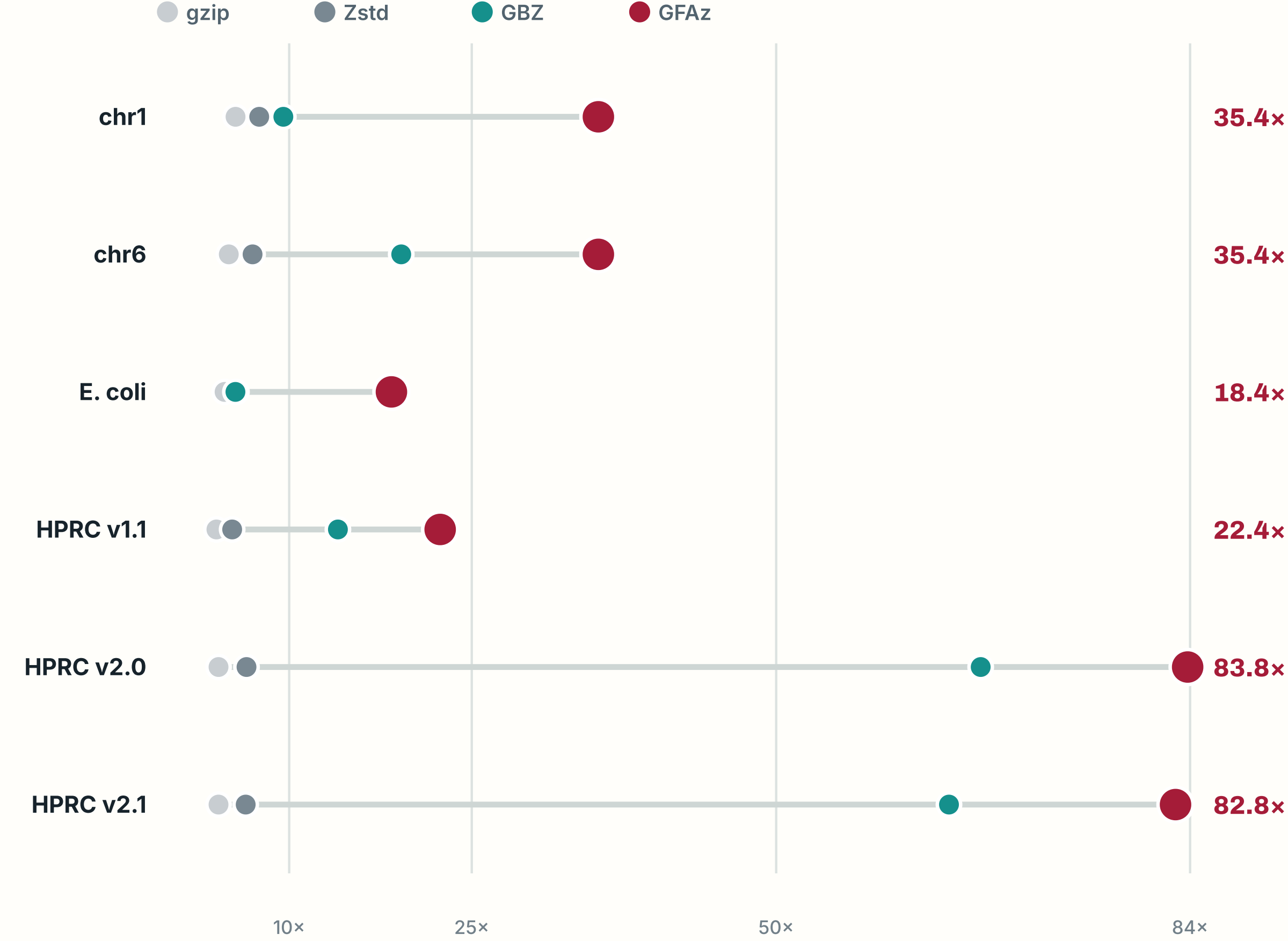
+12 +13 +21 +12 +13 -8 → R₁ +21 R₁ -8
R₁ → (+12,+13) and -R₁ → (-13,-12): one rule represents both orientations. Later rounds combine nodes and rules into longer shared phrases.

- Columnar and lossless
Every GFA record type and optional field is preserved with a type-specific encoding.
- Linear-time grammar
Frequent adjacent pairs are canonicalized across forward and reverse-complement orientation.
- One portable container
The CPU path serializes a shared .gfaz schema; queries can touch only the columns they need.

03 Best ratio across every graph

Compression ratio

higher is better

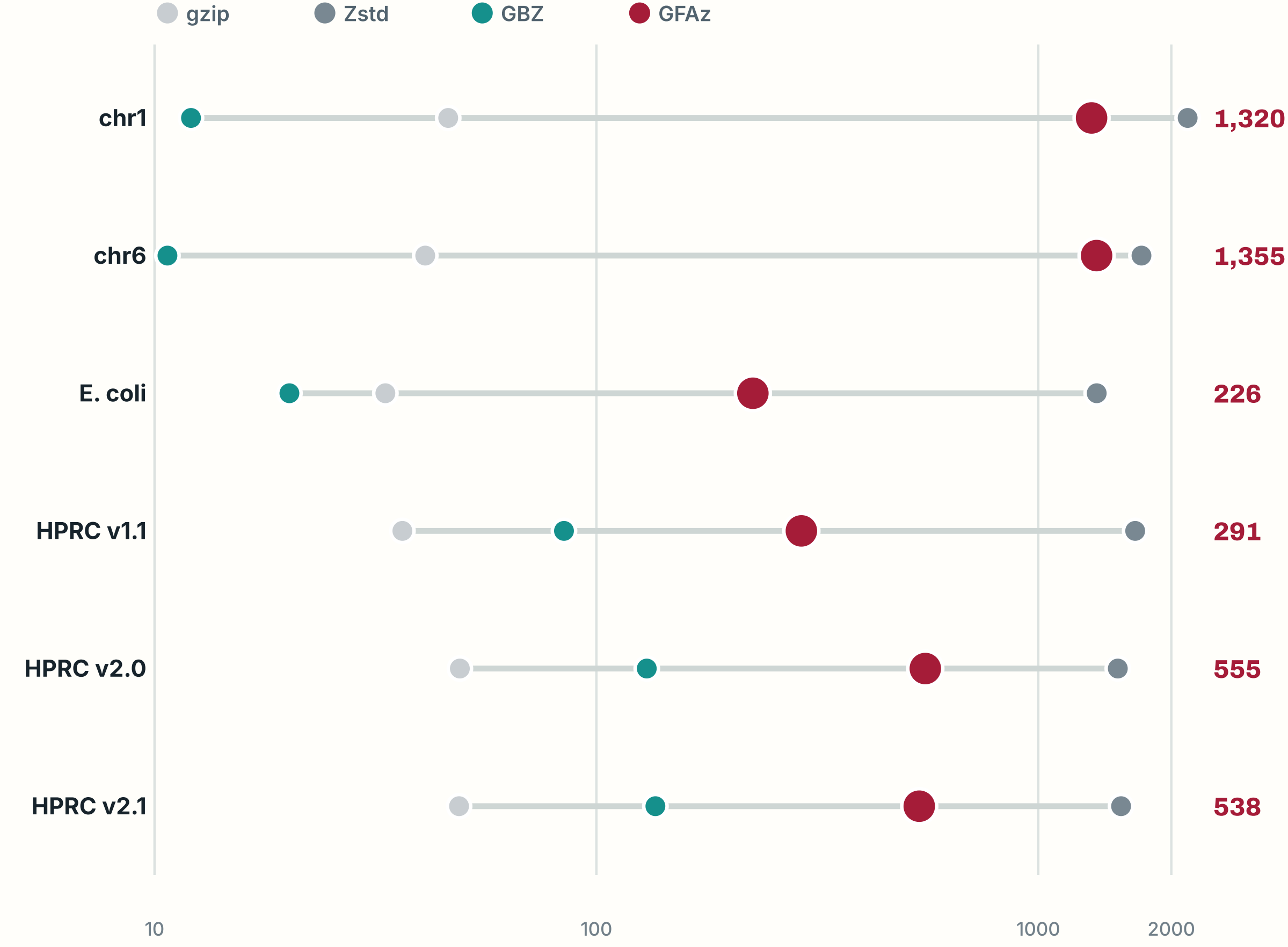


Current repository README data. Ratio is original GFA size divided by compressed size. GFAz preserves full GFA semantics; GBZ preserves a graph/haplotype index rather than lossless GFA text.

04 CPU compression speed

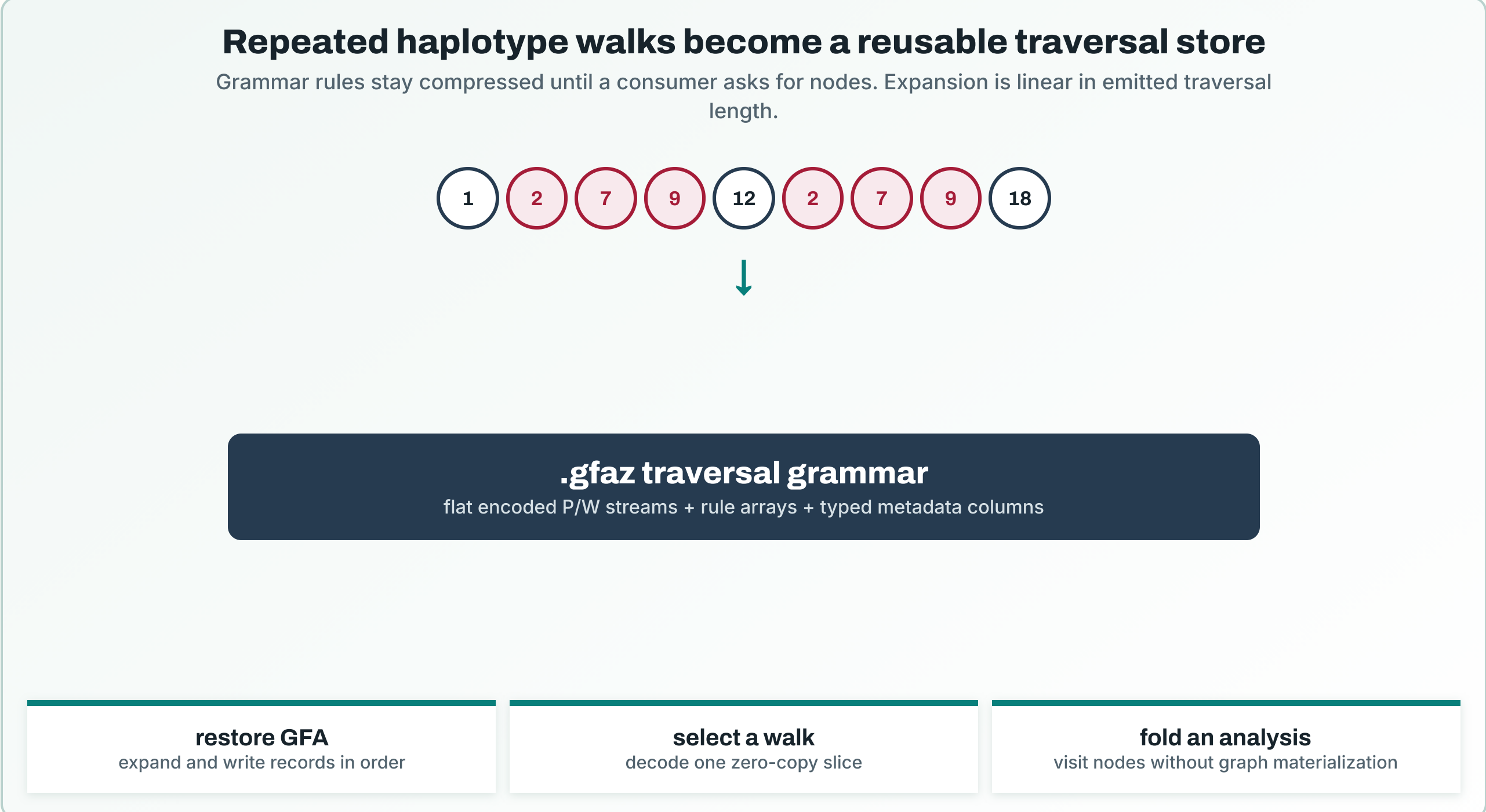
Compression throughput

MiB/s · end-to-end CPU wall time · log scale

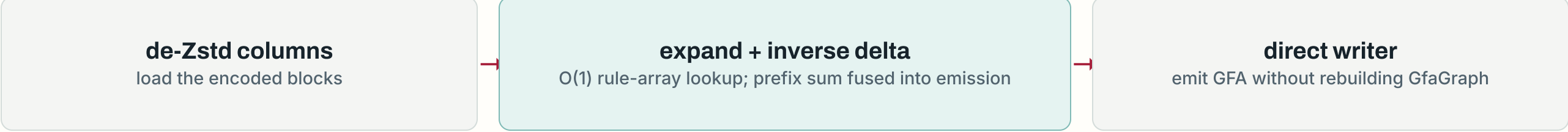


32 threads. End-to-end CLI wall time, including input parsing, grammar construction, entropy encoding, and serialization. Inputs are staged from NVMe immediately before each run.

05 One representation, two streaming paths



06 Streaming decompression avoids reconstruction

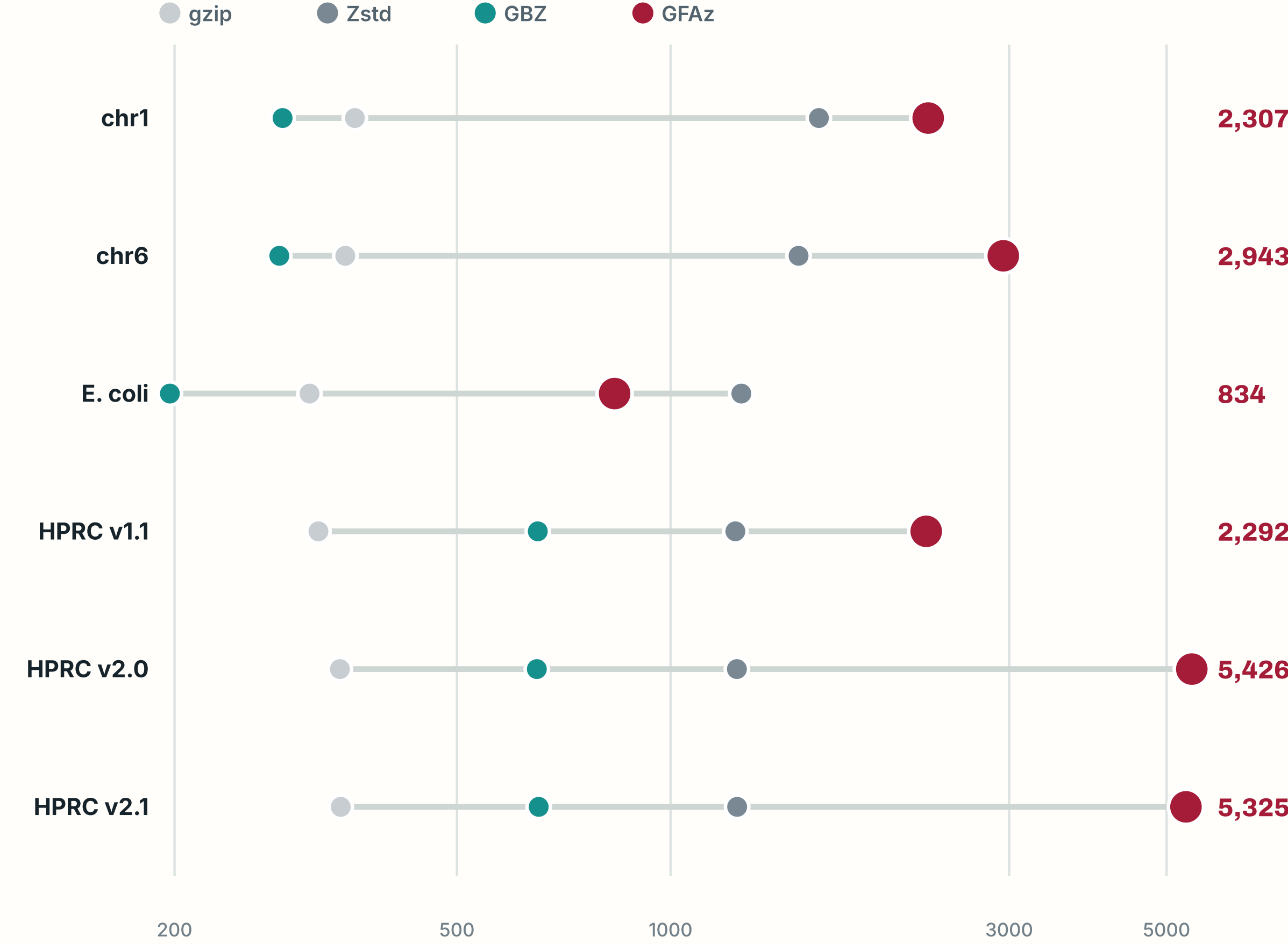


The default CPU decoder uses thread-local output buffers and ordered writes. Fixed-width fields are restored from columns while traversal nodes stream through the grammar expander.

07 CPU decompression speed

Decompression throughput

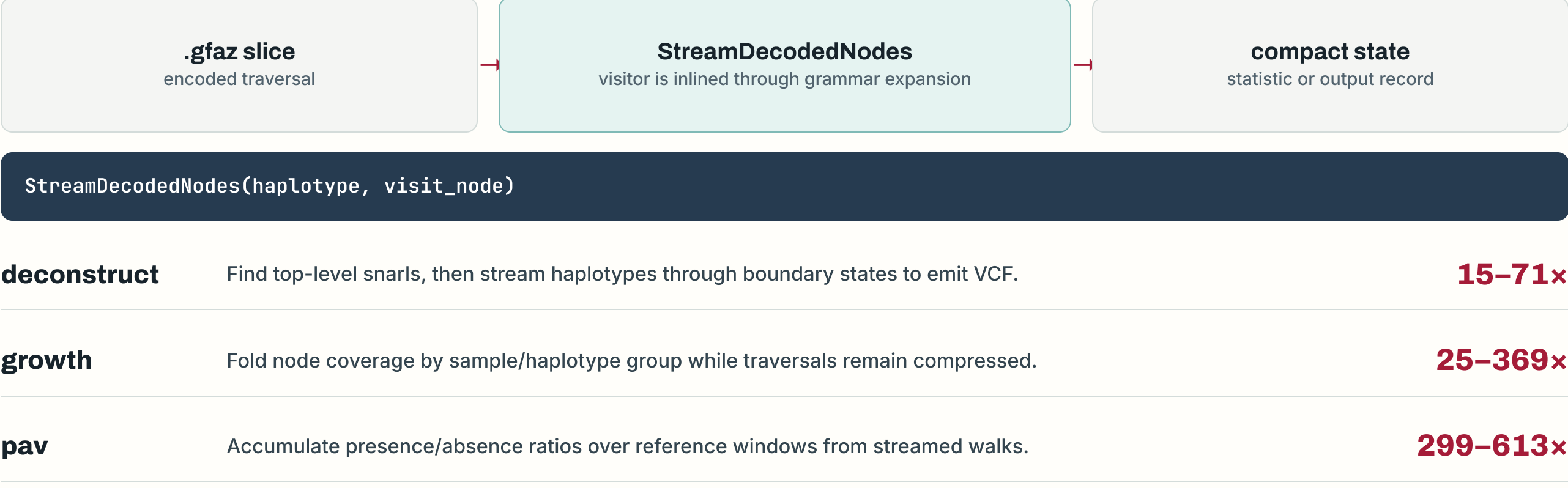
MiB/s · CPU streaming direct writer · log scale



Inputs are preloaded into page cache; decompression writes to /dev/null to isolate compute throughput. GFAz is fastest on five of six datasets and reaches 5.4 GiB/s on HPRC v2.0.

08 The compressed file is also the compute engine

Most path-centric analyses need a fold over haplotypes—not a fully materialized graph.



Speedups are against vg deconstruct, Panacus, and odgi pav at 16 threads.

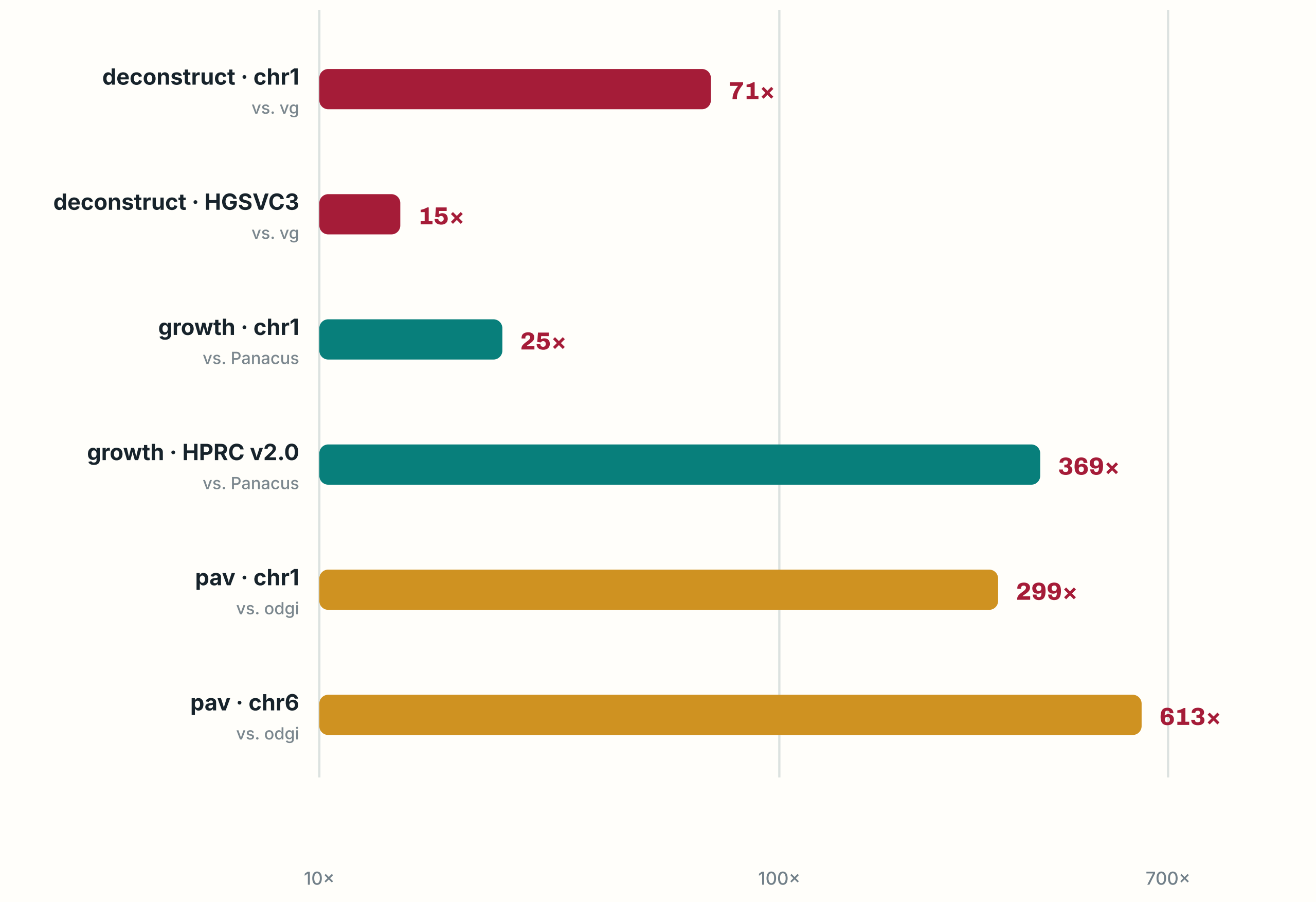
09 The answers agree

~99.99% deconstruct position concordance with vg; record counts within ~0.2%	identical growth curves at every Panacus point
same shape PAV matrices reproduce odgi's reference-window semantics	thread stable byte-identical VCF output at 1 and 16 threads

10 Computing over compression pays off

Direct-analysis speedup

16 CPU threads · log scale · baseline = 1×



Current README comparisons at 16 CPU threads. Baselines read uncompressed GFA; GFAz reads the .gfaz container directly.

11 Whole-genome scale changes what is possible

HPRC v2.0 growth 245 min Panacus · 327 GB RSS	39.8 s GFAz · 12.9 GB RSS
---	-------------------------------------

GFAz reads a ~4.5 GB container and reproduces the exact growth curve. On the largest graphs, vg and odgi cannot load the uncompressed input within the 503 GB node.

12 Current boundary

Whole-genome PAV is not solved yet. Its uncompressed node-set/window index—not traversal decoding—exceeds memory on HPRC v2.0/v2.1. Compressing that working index is the next target.

A pangenome graph does not have to be **unpacked** to be queried.

GFAz turns the largest part of GFA into a representation that is simultaneously compact, fast to decode, and useful for direct analysis.

- 18–84× lossless compression
- up to 5.4 GiB/s CPU decompression
- one to three orders of magnitude faster analysis

